Module 1

Network - group of connected devices

use IP and MAC addresses to make sure communication happens with the right devices

LAN - Local area network
WAN - Wide area network

Network tools

Hub - broadcasts to every device
Switch - makes connections between specific devices
↳ control traffic flow
Router - connects multiple networks
Modem - router to internet

These are physical devices but their features can often be virtualized

Cloud networks

Nowadays more companies use cloud services for networking.

Cloud network - collection of servers or computers that stores resources and data in remote centers accessed via the internet

SaaS - applications
IaaS - physical hardware, servers, storage
PaaS - operating systems, custom environments

Communication

Packet - unit of information sent between devices.

Packets contain header, body, footer

Bandwidth - the amount of data a device receives every second

Sniffing - capturing and inspecting network packets

The TCP/IP model

TCP (transmission control protocol)
IP (internet protocol)

Port - software based location that organizes sending and receiving data

4. Application → How packets interact
3. Transport → Protocols and status
2. Internet → IP addresses assigned, network connection info
1. Link → Physical connections

The OSI model

Application - top level protocol (ftp, smtp, etc)
Presentation - formatting, encryption
Session - authentication, reconnection
Transport - speed, flow
Network - recieve and deliver
Link - sending and recieving
Physical - Physical wiring

IP addresses are unique and identify devices on the internet

IPv4 addresses are x.x.x.x decimal
IPv6 addresses are xxxx:xxxx ... hex

They can be public or private. All on a network share the same public IP. A MAC address is assigned to each local device and used for internal communication

IPv4 Packets

- max size 65,535 bytes
- the header contains the following info:
  - ↳ version VER
  - ↳ IP header length HLEN / IHL
  - ↳ Type of Service TOS
  - ↳ Total Length
  - ↳ Identification
  - ↳ Flags
  - ↳ Fragmentation offset
  - ↳ Time to live TTL
  - ↳ Protocol
  - ↳ Header checksum
  - ↳ Source IP
  - ↳ Dest IP
  - ↳ Options

IPv6 packets are considerably simpler.

Module 2

Protocols

Sets of rules that describe how data should be handled when sending/receiving.

Address resolution protocol (ARP) - used to determine the MAC address of the next router or device

Communication protocols
Management protocols
Security protocols

IEEE 802.11

Wifi protocols

WPA Security protocol for connecting over wifi

WEP -> WPA -> WPA2 -> WPA3

enterprise   personal

Firewalls and Proxies

Network security device that monitors traffic and can block or allow it through the network.

Port filtering - block ports that are uncommon to prevent unwanted communication

There are software and hardware firewalls. Software ones are generally cheaper. Companies may use cloud firewalls

Stateful - firewalls analyse and block
Stateless - uses rules to accept or block
-> considered less secure

NGFW - next-gen firewall even more secure than stateful
-> can use deep packet inspection.

VPN - changes public IP address by encapsulating traffic

Zone

Segment of a network, protecting the internal network

networks can be divided into subnets

Uncontrolled zone - outside networks, not controlled by org
Controlled zone - owned by organisation
-> DMZ, public facing servers
-> internal network, private servers
-> Restricted zone

uncontrolled zone
internet -> Firewall -> DMZ -> Firewall
↓
internal network   controlled zone
↓
Firewall
↓
Restricted zone

Classless inter-domain routing (CIDR) is a method of assigning masks to IP addresses to create a subnet.

CIDR IP addresses are formatted like IPv4 addresses with a / and a number at the end.

Proxy servers fulfil client requests by forwarding them to other servers. Proxy servers use temporary memory for commonly requested resources,

Forward proxy - regulates and restricts access to the internet.

Reverse proxy - restricts internet access to internal servers.

Module 3

Denial of Service attack

disrupt a network by flooding it with traffic to overwhelm the network or prevent legitimate users being able to access it.

a DDoS attack uses multiple devices to target the network.

SYN flood attack – flood a network with handshake requests to block ports

ICMP flood – repeatedly send ICMP packet to get responses.

Ping of death – oversized ICMP packet can crash a system

Packet sniffing

The practice of capturing and analysing packets going across the network.

Passive – packets are read in transit
active – Packets are recorded and then forwarded. They may be altered

IP spoofing

attacker changes the source IP of a system to pretend they are someone they are not.

On-path attack – attacker puts himself in between two devices to read traffic

Replay attack – attacker delays or replays a packet at a later time

Smurf attack – Combination of DDoS and spoofing.

Module 4

hardening is the process of reducing a systems attack surface to prevent intrusion or compromise

Hardening is conducted on:
- OSes
- applications
- hardware
- servers
- databases

Pen-test - simulated attack to find vulnerabilities

Configuration testing - analysing system configurations to find potential vulnerabilities

OS hardening

The OS is the interface between computer hardware and the user.

Patch update - a update that aims to resolve vulnerabilities

Something you have
Something you knew
Something you are

Network hardening

- Port filtering
- Privileges
- encryption
- Path updates
- rule analysis
- log analysis

You should use a SIEM.

Networks should be segmented

Cloud hardening

Consider IAM, configurations and the current attack surface

Note shared responsibility

hypervisors are in two types:
Type 1 - on the hardware
Type 2 - on the software